ANTI-SUBMARINE WARFARE

DURING THE BATTLE of the Atlantic from 1939 to 1945, German U-boats sent around 2,000 Allied merchant vessels to the bottom of the ocean. The U-boat war was the critical threat against Britain once the immediate danger of a German invasion had passed in 1940—Winston Churchill once commented that "the only thing that ever really frightened me during the war was the U-boat peril." Yet over the next six years, the Allies steadily gained the tactical and technological advantage, and by war's end nearly 80 percent of operational U-boats were destroyed, giving the U-boat crews the highest percentage of fatalities of any German armed service.





AIR ATTACK U-boats were acutely vulnerable to surprise air attack when they surfaced. Here a U-boat is strafed by an Allied aircraft, receiving multiple hits to the conning tower.

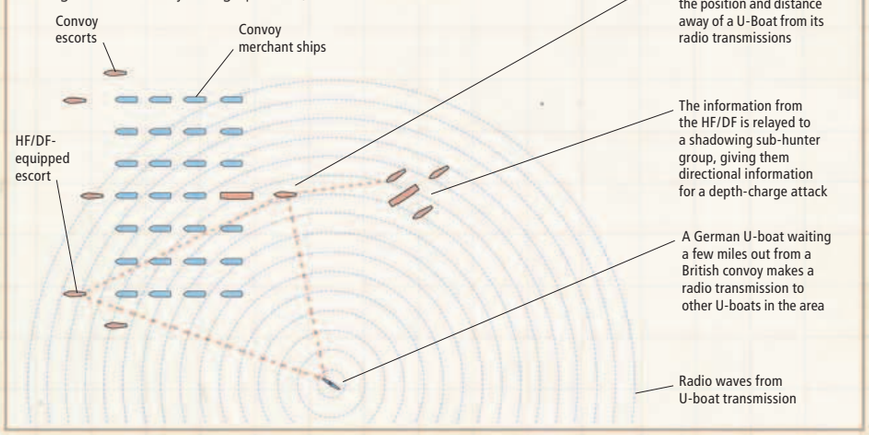
ZIGZAG CLOCK Each convoy ship would have a clock synchronized to sound a bell at irregular intervals, instructing the ships when to change course with the rest of the convoy, helping evade pursuing U-boats.

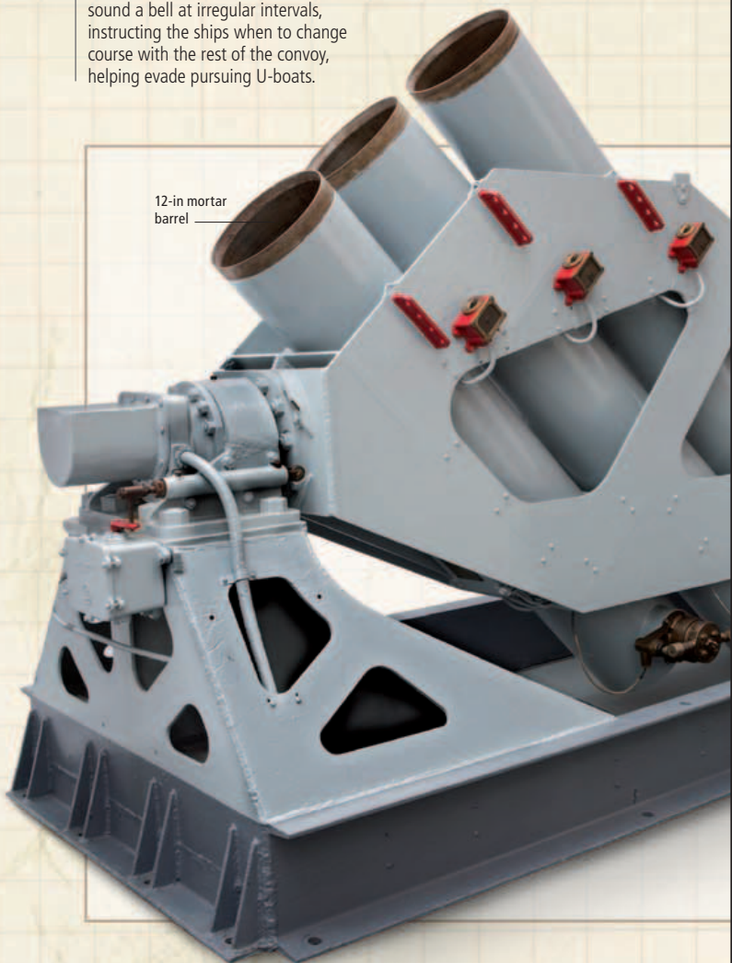
AVOIDING U-BOATS

The primary measure adopted to avoid U-boats was the convoy system. Sending supplies in large convoys rather than single, independent vessels worked on probability. If a U-boat failed to detect a single ship, or failed in an attack, then probability assured that another target would be along soon. However, if the U-boat failed to detect or attack a convoy, then it would lose a mass of opportunities in one instance. Convoying certainly reduced causalities, but the convoys themselves took further evasive measures to throw off the U-boats. They adopted zigzag movements across the Atlantic, with slow convoys making shifts at angles of 20–40 degrees every few minutes. Convoy evasion did not always work, but on many occasions convoys of almost 200 vessels crossed the Atlantic perfectly safely without being spotted by U-boats.

HUFF-DUFF

High-Frequency Direction-Finding (HF/DF), abbreviated to "huff-duff," became one of the crucial technological resources in the battle against the U-boats. It consisted of either shore-based or ship-mounted antennas that picked up German U-boat radio transmissions, from which it was possible to triangulate the enemy's rough position, and then make evasion or attack.





LOCATING U-BOATS

Even more than "huff-duff" (see above), radar systems were the critical U-boat locating tool. The first ship-mounted sets were crude and often ineffective, but by 1942 the British Type 271 centimetric radar was in operation, with enough sensitivity to detect a surfaced submarine at a distance of several miles. Once similar radar was fitted to long-range patrol aircraft, U-boats found it more difficult to hide. When the submarine was submerged, ASDIC (sonar) came into play. ASDIC used a sonar pulse that, if it hit and rebounded off a submarine's hull, would give depth and direction data. The Allies also used hydrophone sets, powerful underwater microphones that "listened" for propeller or other noises. The Americans had the Magnetic Anomaly Device (MAD) fitted to patrol aircraft, which searched for magnetic disturbances in the water.